

Image Enhancement in the Frequency Domain

Subtopic 3.3: Smoothing Frequency Domain Filters

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Outline - Subtopic 3.3: Smoothing Frequency Domain Filters

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Frequency Domain Filtering: Overview

- **Core Concept:** Modify frequency components to enhance or suppress specific features
- **General Process:**
 - ① Compute 2-D DFT of image
 - ② Multiply by filter function $H(u, v)$
 - ③ Compute inverse DFT to get filtered image

- **Mathematical Expression:**

$$G(u, v) = H(u, v) \cdot F(u, v) \quad (1)$$

where $G(u, v)$ is filtered spectrum, $F(u, v)$ is original spectrum

- **Why Low-Pass (Smoothing) Filters?**
 - Reduce high-frequency noise
 - Blur image to smooth transitions
 - Remove sharp details while preserving overall structure

Low-Pass Filters: Purpose and Characteristics

- **Objective:** Allow low frequencies to pass, attenuate high frequencies
- **Effect on Image:**
 - Smoothing and blurring
 - Noise reduction
 - Edge softening
 - Loss of fine details
- **Key Parameter:** Cutoff frequency D_0
 - Determines transition between passing and attenuating frequencies
 - Controls degree of smoothing
 - Related to distance from origin in frequency domain
- **Three Main Types to Study:**
 - 1 Ideal Low-Pass Filter (ILPF)
 - 2 Butterworth Low-Pass Filter (BLPF)
 - 3 Gaussian Low-Pass Filter (GLPF)

Ideal Low-Pass Filter: Definition

- **Mathematical Definition:**

$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{if } D(u, v) > D_0 \end{cases} \quad (2)$$

where $D(u, v)$ is distance from origin in frequency domain

- **Distance Function:**

$$D(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2} \quad (3)$$

- **Cutoff Frequency:** D_0 determines filter radius
 - Sharp transition at cutoff
 - Frequencies within D_0 : pass completely
 - Frequencies beyond D_0 : blocked completely

ILPF: Characteristics and Properties

- **Advantages:**

- Simplest to understand and implement
- Sharp frequency cutoff
- Precise control at cutoff frequency

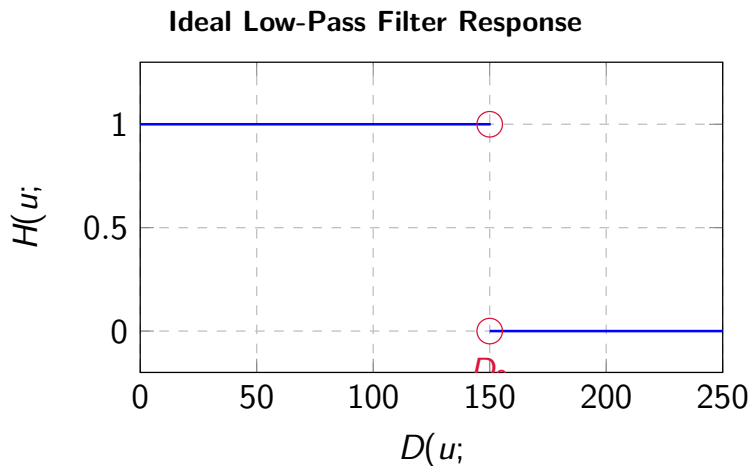
- **Disadvantages:**

- **Ringings Artifact:** Introduces oscillations around edges
- Causes “halo” effect at sharp transitions
- Poor in practice due to Gibbs phenomenon
- Abrupt discontinuity creates spatial domain artifacts

- **Gibbs Phenomenon:**

Sharp transition in frequency $\Rightarrow$ Ringing in spatial domain (4)

ILPF: Visual Representation



Perfect cutoff at D_0 but causes ringing artifacts

ILPF: Example Application

- **Example:** Image size 256×256 , $D_0 = 30$
- **Center at:** (128, 128) (using frequency domain centering)
- **Frequency Response:**
 - All frequencies within circular region of radius 30: **pass**
 - All frequencies outside: **blocked**
- **Visual Result:**
 - Strong smoothing and blur
 - Visible ringing around edges
 - “Halo” artifacts around high-contrast areas
 - Loss of edge sharpness
- **Use Case:** Limited practical use; mainly theoretical reference

Butterworth Low-Pass Filter: Definition

- **Mathematical Definition:**

$$H(u, v) = \frac{1}{1 + \left[\frac{D(u, v)}{D_0} \right]^{2n}} \quad (5)$$

where $D(u, v)$ is distance and n is filter order

- **Key Parameters:**

- D_0 : Cutoff frequency (distance at which $H = 0.5$)
- n : Filter order (controls transition sharpness)

- **At Cutoff Frequency ($D = D_0$):**

$$H(D_0, v) = \frac{1}{1 + 1} = 0.5 \quad (-3 \text{ dB point}) \quad (6)$$

BLPF: Order Effect on Response

- **Order Behavior:**

- $n = 1$: Very smooth transition, gradual rolloff
- $n = 2$: Moderate transition
- $n = 4$: Sharper transition approaching ideal filter
- $n \rightarrow \infty$: Approaches ideal low-pass filter

- **Transition Characteristics:**

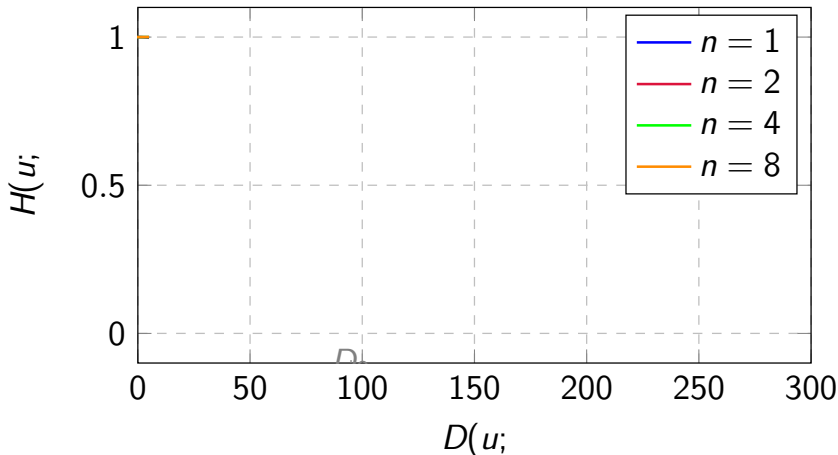
- No sharp cutoff like ILPF
- Smooth, monotonic decrease
- Less ringing than ILPF
- Better practical performance

- **Filter Shape:**

- Circular response in 2-D frequency domain
- Radially symmetric
- Same behavior in all directions

BLPF: Visual Comparison for Different Orders

Butterworth Low-Pass Filter for Various Orders



Higher order approaches ideal filter behavior

BLPF: Advantages and Properties

- **Advantages:**

- Smooth transition (no sharp discontinuity)
- Minimal ringing artifacts
- Adjustable transition with order parameter
- Practical and widely used
- Monotonic rolloff

- **Trade-offs:**

- Higher order $\Rightarrow$ sharper cutoff but more computation
- Lower order $\Rightarrow$ smoother but less selective
- Order $n = 2$ often good compromise

- **Typical Applications:**

- Image smoothing and noise reduction
- Preprocessing for feature extraction
- General-purpose low-pass filtering

- At Key Distances:

$$D = 0 : H(0, 0) = 1 \quad (\text{DC component passes}) \quad (7)$$

$$D = D_0 : H(D_0, D_0) = 0.5 \quad (-3 \text{ dB}) \quad (8)$$

$$D = 2D_0 : H(2D_0, 2D_0) = \frac{1}{1 + 2^{2n}} \quad (\text{small value}) \quad (9)$$

- Order Effect on Attenuation:

$$H(2D_0, \nu) = \begin{cases} 0.2 & \text{if } n = 1 \\ 0.039 & \text{if } n = 2 \\ 0.0015 & \text{if } n = 4 \\ 0.00006 & \text{if } n = 8 \end{cases} \quad (10)$$

Gaussian Low-Pass Filter: Definition

- **Mathematical Definition:**

$$H(u, v) = e^{-D^2(u, v)/(2D_0^2)} \quad (11)$$

where $D(u, v)$ is distance from origin, D_0 is cutoff frequency

- **Alternative Form:**

$$H(u, v) = e^{-\sigma^2 D^2(u, v)/2} \quad (12)$$

where σ controls the standard deviation (width)

- **Characteristics at Cutoff:**

$$H(D_0, D_0) = e^{-1/2} \approx 0.6065 \quad (\text{not } -3 \text{ dB like Butterworth}) \quad (13)$$

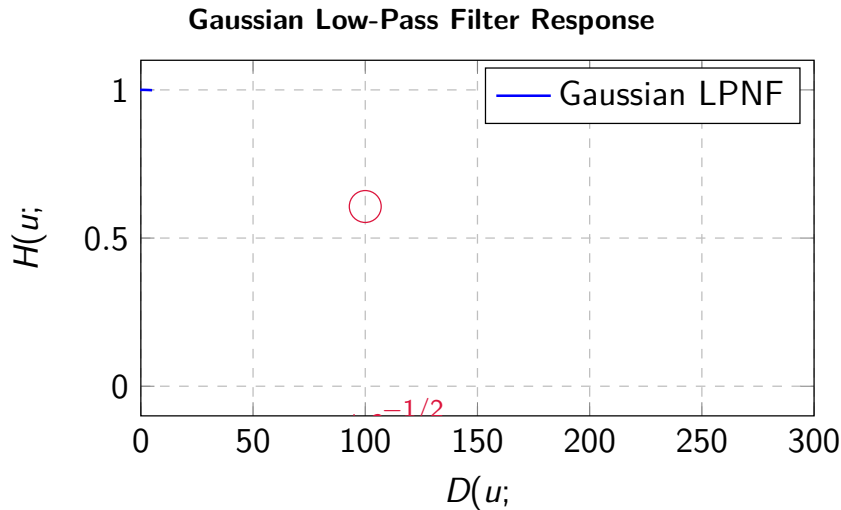
GLPF: Properties and Characteristics

- **Key Properties:**
 - **Smooth:** No sharp cutoff, smooth Gaussian curve
 - **No Ringing:** Spatial domain has no oscillations
 - **Minimal Artifacts:** Very smooth transitions
 - **Symmetry:** Both frequency and spatial domain are Gaussian
- **Relationship to Butterworth:**
 - Smoother than Butterworth at all orders
 - No ringing unlike Butterworth at high orders
 - Easier to implement
- **Spatial Domain Equivalence:**

Gaussian in frequency $\Leftrightarrow$ Gaussian in spatial domain (14)

(Due to Fourier transform property)

GLPF: Visual Representation



Smooth, bell-shaped response with no sharp cutoff

GLPF: Advantages and Applications

- **Advantages:**

- **No Ringing:** Smoothest possible transition
- **Best Quality:** Most natural-looking results
- **Mathematical Elegance:** Self-dual in frequency and spatial domains
- **Easy to Use:** Simple parameter (D_0)
- **Stable:** Numerically stable computation

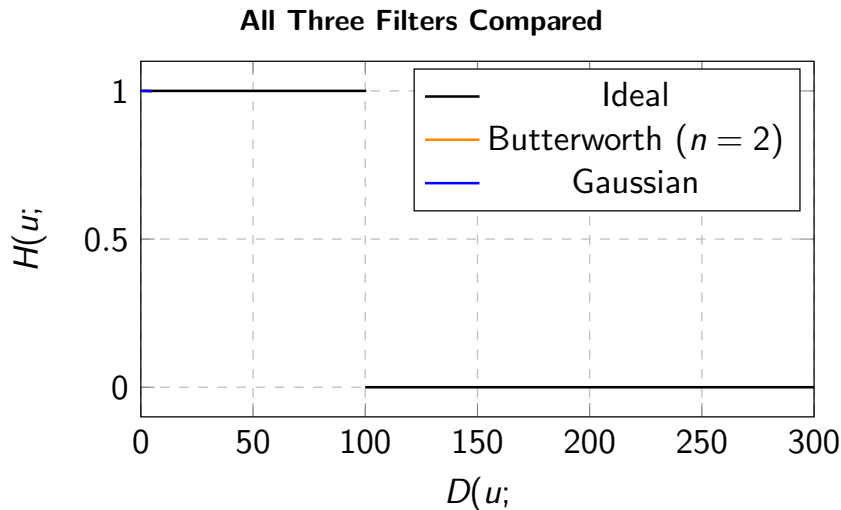
- **Applications:**

- Image smoothing with minimal artifacts
- Pre-processing for computer vision
- Noise reduction in sensitive applications
- Medical image processing
- High-quality image enhancement

Comparison: ILPF vs BLPF vs GLPF

Aspect	ILPF	BLPF	GLPF
Formula	Sharp cutoff	$(1 + [D/D_0]^{2n})^{-1}$	$e^{-D^2/(2D_0^2)}$
Cutoff	Abrupt	At -3 dB	At $e^{-1/2}$
Transition	Discontinuous	Smooth	Smoothest
Ringings	Severe	Moderate	None
Artifacts	Halo effects	Minimal	Minimal
Complexity	Low	Medium	Low
Parameters	1 (D_0)	2 (D_0, n)	1 (D_0)
Practical Use	Theory only	Common	Preferred

Frequency Response Comparison



Gaussian has smoothest response; Ideal has sharp discontinuity

Implementation: Practical Steps

- **Step 1: Prepare Image**

- Pad image to avoid circular convolution
- Typically pad to power of 2 for FFT efficiency

- **Step 2: Compute 2-D DFT**

$$F(u, v) = \text{FFT2D}(f(x, y)) \quad (15)$$

- **Step 3: Center the Spectrum**

$$F_c(u, v) = F(u, v) \cdot (-1)^{u+v} \quad (16)$$

- **Step 4: Apply Filter**

$$G_c(u, v) = H(u, v) \cdot F_c(u, v) \quad (17)$$

- **Step 5: Inverse Transform and Extract Real Part**

Distance Calculation from Center

- **Standard Centering:** Origin at $(0,0)$

$$D(u, v) = \sqrt{u^2 + v^2} \quad (19)$$

- **Centered Coordinate System:** Origin at $(M/2, N/2)$

$$D(u, v) = \sqrt{(u - M/2)^2 + (v - N/2)^2} \quad (20)$$

- **Implementation Tips:**

- Use centered coordinates for intuitive filter design
- Use `fftshift` in MATLAB/Python for centering
- Precompute distance matrix for efficiency

- **Computational Efficiency:**

- Cache distance matrix: $D[i, j]$ pre-computed
- Avoid recomputation in filter loops
- Use vectorized operations

Choosing Filter Parameters

- **Cutoff Frequency D_0 :**
 - **Small D_0** ($\approx 10 - 30$): Heavy smoothing, blur
 - **Medium D_0** ($\approx 50 - 100$): Moderate smoothing
 - **Large D_0** ($\approx 100 - 200$): Gentle smoothing
 - **Guidelines:** Typically 5% to 10% of image dimension
- **Butterworth Order n :**
 - $n = 1$: Smooth, weak filtering
 - $n = 2$: Good balance (most common)
 - $n = 4$: Sharper cutoff, more artifacts
 - Higher n : Approaching ideal filter
- **Selection Strategy:**
 - Start with Gaussian for best results
 - If more control needed, use Butterworth with $n = 2$
 - Avoid Ideal filter in practice

Visual Effects of Low-Pass Filtering

- **Smoothing Effects:**

- Blurs sharp transitions
- Reduces fine details
- Softens edges
- Reduces high-frequency noise

- **Ringling Artifacts:**

- Most severe in Ideal filter
- Moderate in Butterworth (high order)
- Absent in Gaussian filter
- Visible as oscillations near edges

- **Loss of Information:**

- Higher frequencies removed
- Edge sharpness reduced
- Finer structural details lost
- Trade-off with noise reduction

Practical Applications

- **Noise Reduction:**
 - Remove high-frequency noise components
 - Preserve low-frequency image structure
 - Gaussian filter best for noise with fine structure
- **Image Pre-processing:**
 - Prepare image for edge detection
 - Normalize intensity variations
 - Reduce texture artifacts
- **Blurring Effects:**
 - Create soft-focus effects
 - Blur background in image composition
 - Artistic image processing
- **Medical Imaging:**
 - Reduce scanner noise
 - Smooth speckle artifacts
 - Improve image quality for analysis

Algorithm: Low-Pass Filtering in Frequency Domain

[H] Frequency Domain Low-Pass Filtering **Input:** Image $f(x, y)$, Filter type, Cutoff D_0

Output: Filtered image $g(x, y)$ 1. Pad image to $(2M) \times (2N)$ or nearest power of 2 2.

Compute 2-D DFT: $F(u, v) \leftarrow \text{FFT2D}(f)$ 3. Center spectrum:

$F_c(u, v) \leftarrow F(u, v) \cdot (-1)^{u+v}$ 4. Create distance matrix: $D[i, j] \leftarrow \sqrt{(i - M)^2 + (j - N)^2}$

5. **Compute filter** $H[i, j]$: **if** Filter = ILPF: $H[i, j] \leftarrow (D[i, j] \leq D_0) ? 1 : 0$ **elseif**

Filter = BLPF (order n): $H[i, j] \leftarrow 1 / (1 + [D[i, j] / D_0]^{2n})$ **elseif** Filter = GLPF:

$H[i, j] \leftarrow e^{-D[i, j]^2 / (2D_0^2)}$ 6. Apply filter: $G_c(u, v) \leftarrow H(u, v) \times F_c(u, v)$ 7. Inverse transform:

$g(x, y) \leftarrow \text{Real}(\text{IFFT2D}(G_c))$ 8. **Return** $g(x, y)$

Summary Table: Low-Pass Filters

Filter	Equation	Best For	Issues
Ideal	$H = \begin{cases} 1 & D \leq D_0 \\ 0 & D > D_0 \end{cases}$	Theory	Severe ringing
Butterworth	$H = \frac{1}{1 + [D/D_0]^{2n}}$	Control	Order tuning
Gaussian	$H = e^{-D^2/(2D_0^2)}$	Practice	Slower rolloff

Characteristic	ILPF	BLPF	GLPF
Ringing Artifacts	Severe	Moderate	None
Implementation	Simple	Medium	Simple
Parameters	1	2	1
Practical Use	No	Yes	Preferred
Quality	Poor	Good	Excellent

Textbook and Reference Materials

Recommended Reading

- **Textbook:**

- Rafael C. Gonzalez and Richard E. Woods, *Digital Image Processing*, Pearson Edition, Latest Edition.

- **Reference Books:**

- I. Pitas, *Digital Image Processing Algorithms*, Prentice Hall, Latest Edition.
- A. K. Jain, *Fundamentals of Digital Image Processing*, Prentice Hall of India Pvt. Ltd., Latest Edition.
- K. Castleman, *Digital Image Processing*, Prentice Hall of India Pvt. Ltd., Latest Edition.

- **Related Topics:**

- High-pass filters (sharpening)
- Band-pass filters (selective frequency enhancement)
- Notch filters (interference removal)

Key Takeaways - Smoothing Filters

- **Ideal Low-Pass Filter:** Theoretical reference; impractical due to severe ringing
- **Butterworth Low-Pass Filter:** Good balance; adjustable with order parameter
- **Gaussian Low-Pass Filter:** Best practical choice; smooth, artifact-free
- **Cutoff Frequency:** Controls degree of smoothing
- **Trade-offs:** Balance between noise reduction and detail preservation
- **Implementation:** Frequency domain filtering is computationally efficient

Practical Recommendation:

Use Gaussian filter for best results;
Butterworth for tunable filtering